

CSE 222 - Homework #7

Balanced Tree-Based Stock Data Management

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Key Methods and Their Functionality

- AVLTree Class

- 1. Node Class**

- Stores stock data and pointers to left and right child nodes.
 - Contains a height attribute to manage tree balance.

- 2. insert(Stock):**

- Inserts a new stock or updates an existing stock in the tree.
 - Balances the tree if necessary after insertion.

- 3. delete(String)**

- Deletes a stock based on its symbol.
 - Balances the tree if necessary after deletion.

- 4. search(String)**

- Searches for a stock by its symbol and returns the stock data if found.

- 5. rotateLeft(Node) & rotateRight(Node)**

- Performs left and right rotations to balance the tree.

- 6. inOrderTraversal(), preOrderTraversal(), postOrderTraversal()**

- Provides different tree traversal methods for debugging and validation.

- StockDataManager Class

- 1. addOrUpdateStock(String, double, long, long)**

- Adds a new stock or updates an existing stock.

- Uses AVLTree's insert method to manage the stock data.

2. removeStock(String)

- Removes a stock by its symbol.
- Uses AVLTree's delete method.

3. searchStock(String)

- Searches for a stock by its symbol.
- Uses AVLTree's search method.

4. updateStock(String, String, double, long, long)

- Updates a stock's details, including its symbol, price, volume, and market cap.

- **RandomInputGenerator Class**

- To facilitate testing, a random input generator was created. This generator produces a file with random commands for adding, updating, searching, and removing stocks.

Handling Tree Balancing

The AVL Tree maintains its balance by ensuring that the difference in height between the left and right subtrees of any node is at most one. This is achieved through rotations during insertion and deletion:

- Left Rotation: Performed when a node's right subtree is heavier.
- Right Rotation: Performed when a node's left subtree is heavier.
- Left-Right Rotation: Performed when a node's left child has a right-heavy subtree.
- Right-Left Rotation: Performed when a node's right child has a left-heavy subtree.

Input File Format

The input file contains a series of commands to manipulate the AVL Tree. Each command is on a new line, and the format is as follows:

- `ADD <symbol> <price> <volume> <marketCap>`: Adds or updates a stock.

- ``REMOVE <symbol>``: Removes a stock.
- ``SEARCH <symbol>``: Searches for a stock.
- ``UPDATE <symbol> <newSymbol> <newPrice> <newVolume> <newMarketCap>``: Updates a stock's details.

Performance Analysis

Performance analysis was conducted 1000 times for each operation (ADD, REMOVE, SEARCH, UPDATE) to validate the time complexity of the AVL Tree implementation. The operations were measured and plotted to show the "O(log n)" complexity.

To be able to see the performance results we need to uncomment this line in the Main.java:

```
performPerformanceAnalysis(manager, 1000);
```

Challenges & Solutions

- Balancing After Insertions/Deletions: Handled through appropriate rotations.
- Duplicate Symbols: Managing duplicate stock symbols required updating the stock's data instead of inserting a new node. This was addressed by checking for existing symbols before insertion.

Compiling

To change the input file we need to change the file name in the "run" command from the makefile. For testing, there is two input files in the "inputFiles" directory.

To compile the project follow these steps:

<code># make</code>	--> compiles all classes
<code># make generate</code>	--> generates a random input file
<code># make run</code>	--> runs the project with the generated file as a default
<code># make doc</code>	--> creates JavaDoc for the project

Sample Scenarios

```

makefile  input2.txt  Main.java
1  ADD AAPL 150.0 100000 2500000000
2  ADD GOOGL 2800.0 500000 1500000000
3  ADD MSFT 300.0 800000 2000000000
4  ADD AMZN 3400.0 300000 1700000000
5  ADD TSLA 700.0 600000 7000000000
6  ADD FB 270.0 900000 1800000000
7  ADD NFLX 500.0 400000 1000000000
8  ADD NVDA 650.0 700000 1400000000
9  ADD BABA 220.0 500000 1100000000
10 ADD V 230.0 200000 500000000
11 SEARCH AAPL
12 SEARCH GOOGL
13 SEARCH MSFT
14 SEARCH AMZN
15 SEARCH TSLA
16 UPDATE AAPL APPL 155.0 1050000 2550000000
17 UPDATE GOOGL GOOG 2850.0 550000 1550000000
18 UPDATE MSFT MSFT 310.0 850000 2050000000
19 UPDATE AMZN AMZ 3450.0 320000 1750000000
20 UPDATE TSLA TSLA 720.0 620000 7200000000
21 REMOVE FB
22 REMOVE NFLX
23 REMOVE NVDA
24 REMOVE BABA
25 REMOVE V
26 ADD TWTR 65.0 150000 320000000
27 ADD SNAP 55.0 130000 310000000
28 ADD PINS 60.0 170000 340000000
29 ADD SQ 200.0 180000 600000000
30 ADD PYPL 250.0 220000 880000000
31 ADD SHOP 1200.0 100000 1200000000
32 ADD SPOT 250.0 90000 500000000

PS C:\Workspace\2023-spr\cse222\hw07\HW7> make run
java Main inputFiles/input2.txt
Symbol AAPL added!
Symbol GOOGL added!
Symbol MSFT added!
Symbol AMZN added!
Symbol TSLA added!
Symbol FB added!
Symbol NFLX added!
Symbol NVDA added!
Symbol BABA added!
Symbol V added!
Stock [symbol=AAPL, price=150.0, volume=1000000, marketCap=2500000000]
Stock [symbol=GOOGL, price=2800.0, volume=500000, marketCap=1500000000]
Stock [symbol=MSFT, price=300.0, volume=800000, marketCap=2000000000]
Stock [symbol=AMZN, price=3400.0, volume=300000, marketCap=1700000000]
Stock [symbol=TSLA, price=700.0, volume=600000, marketCap=7000000000]
Symbol AAPL updated - new symbol: APPL
Symbol GOOGL updated - new symbol: GOOG
Symbol MSFT updated - new symbol: MSFT
Symbol AMZN updated - new symbol: AMZ
Symbol TSLA updated - new symbol: TSLA
Symbol TWTR added!
Symbol SNAP added!
Symbol PINS added!
Symbol SQ added!
Symbol PYPL added!
Symbol SHOP added!
Symbol SPOT added!
Symbol UBER added!
Symbol LYFT added!
Stock [symbol=TWTR, price=65.0, volume=150000, marketCap=320000000]
Stock [symbol=SNAP, price=55.0, volume=130000, marketCap=310000000]

```

```

2  ADD SPOT 250.0 90000 500000000
3  ADD UBER 45.0 200000 400000000
4  ADD LYFT 60.0 50000 100000000
5  SEARCH TWTR
6  SEARCH SNAP
7  SEARCH PINS
8  SEARCH SQ
9  SEARCH PYPL
10 UPDATE SHOP SHOP 1250.0 105000 1250000000
11 UPDATE SPOT SPOT 255.0 95000 505000000
12 UPDATE UBER UBER 50.0 210000 410000000
13 UPDATE LYFT LYFT 65.0 55000 105000000
14 REMOVE TWTR
15 REMOVE SNAP
16 REMOVE PINS
17 REMOVE SQ
18 REMOVE PYPL

Symbol AMZ updated - new symbol: AMZ
Symbol GOOGL updated - new symbol: GOOG
Symbol MSFT updated - new symbol: MSFT
Symbol AMZN updated - new symbol: AMZ
Symbol TSLA updated - new symbol: TSLA
Symbol TWTR added!
Symbol SNAP added!
Symbol PINS added!
Symbol SQ added!
Symbol PYPL added!
Symbol SHOP added!
Symbol SPOT added!
Symbol UBER added!
Symbol LYFT added!
Stock [symbol=TWTR, price=65.0, volume=150000, marketCap=320000000]
Stock [symbol=SNAP, price=55.0, volume=130000, marketCap=310000000]
Stock [symbol=PINS, price=60.0, volume=170000, marketCap=340000000]
Stock [symbol=SQ, price=200.0, volume=180000, marketCap=600000000]
Stock [symbol=PYPL, price=250.0, volume=220000, marketCap=880000000]
Symbol SHOP updated - new symbol: SHOP
Symbol SPOT updated - new symbol: SPOT
Symbol UBER updated - new symbol: UBER
Symbol LYFT updated - new symbol: LYFT
PS C:\Workspace\2023-spr\cse222\hw07\HW7>

```

makefile		random_input.txt	Main.java	Terminal	Local
1	SEARCH RRC			make: Nothing to be done for 'all'.	
2	UPDATE FUP DAO 728.76 73168684 68803386			PS C:\Workspace\2023-spr\cse222\hw07\HW7> make run	
3	REMOVE QIW			java Main random_input.txt	
4	ADD KID 640.66 8262756 67979730			Stock not found: RRC	
5	REMOVE BMG			Symbol FUP not found!	
6	SEARCH ZRT			Symbol QIW not found!	
7	UPDATE RLL FBM 336.30 58325332 69929590			Symbol KID added!	
8	REMOVE WVS			Symbol BMG not found!	
9	SEARCH OVO			Stock not found: ZRT	
10	REMOVE SJX			Symbol RLL not found!	
11	UPDATE ZHY BFJ 582.69 26683426 65048126			Symbol WVS not found!	
12	REMOVE NUD			Stock not found: OVO	
13	REMOVE OZC			Symbol SJX not found!	
14	REMOVE JKI			Symbol ZHY not found!	
15	SEARCH QRX			Symbol NUD not found!	
16	UPDATE WAW ABP 931.40 50428269 41380878			Symbol OZC not found!	
17	SEARCH KZO			Symbol JKI not found!	
18	SEARCH QOZ			Stock not found: QRX	
19	SEARCH KVB			Symbol WAW not found!	
20	UPDATE ZZG KNZ 977.15 34697295 846055			Stock not found: KZO	
21	REMOVE AAC			Stock not found: QOZ	
22	UPDATE FYJ LAW 74.40 50029307 80059987			Stock not found: KVB	
23	UPDATE KIE LQQ 387.95 48804400 30386189			Symbol ZZG not found!	
24	ADD OCY 27.68 34043061 59936985			Symbol AAC not found!	
25	REMOVE RPM			Symbol FYJ not found!	
26	SEARCH WKB			Symbol KIE not found!	
27	UPDATE FWW KDZ 89.43 3421045 47335197			Symbol OCY added!	
28	REMOVE CSI			Symbol RPM not found!	
29	REMOVE COO			Stock not found: WKB	
30	REMOVE TZD			Symbol FWW not found!	
31	UPDATE RZU IOU 271.91 10302828 86771841			Symbol CSI not found!	
32	ADD FID 352.67 94668154 70156169			Symbol COO not found!	
				Symbol TZD not found!	

	makefile	random_input.txt	Main.java	Terminal
25	REMOVE	RPM		Stock not found: KVB
26	SEARCH	WKB		Symbol ZZG not found!
27	UPDATE	FWW KDZ 89.43 3421045 47335197		Symbol AAC not found!
28	REMOVE	CSI		Symbol FYJ not found!
29	REMOVE	C00		Symbol KIE not found!
30	REMOVE	TZD		Symbol OCY added!
31	UPDATE	RZU IOU 271.91 10302828 86771841		Symbol RPM not found!
32	ADD	FID 352.67 94668154 70156169		Stock not found: WKB
33	REMOVE	LYJ		Symbol FWW not found!
34	SEARCH	HVH		Symbol CSI not found!
35	UPDATE	THA KFE 224.96 13421673 37686022		Symbol C00 not found!
36	SEARCH	WSJ		Symbol TZD not found!
37	ADD	FXA 306.02 15653576 61758041		Symbol RZU not found!
38	REMOVE	GYN		Symbol FID added!
39	UPDATE	GFH JXG 665.13 31771782 95583214		Symbol LYJ not found!
40	ADD	WWX 666.34 17553274 62185005		Stock not found: HVH
41	REMOVE	NYF		Symbol THA not found!
42	SEARCH	ANB		Stock not found: WSJ
43	UPDATE	PQY QKO 827.71 95024692 9463840		Symbol FXA added!
44	REMOVE	EIX		Symbol GYN not found!
45	REMOVE	KVH		Symbol GFH not found!
46	UPDATE	DNX L0B 642.67 40103752 62638176		Symbol WWX added!
47	ADD	BKE 687.59 15250028 95831983		Symbol NYF not found!
48	REMOVE	TOX		Stock not found: ANB
49	REMOVE	NJS		Symbol PQY not found!
50	SEARCH	YUB		Symbol EIX not found!
51				Symbol KVH not found!
				Symbol DNX not found!
				Symbol BKE added!
				Symbol TOX not found!
				Symbol NJS not found!
				Stock not found: YUB
				PS C:\Workspace\2023-spr\cse222\hw07\HW7>

Performance analysis for 1000 inputs:

